1. Clinical Question & Core Hypothesis

Clinical Question

Why do some high-risk localized prostate cancers fail to respond to AR-targeted therapies despite apparent androgen dependence? The HTAV framing suggests that intratumoral microarchitectural and spatial heterogeneity could create niches that enable pre-existing or rapidly emergent AR-independent clones to survive AR blockade. 2) Tier 1 paths used: No ontologist tiers were provided, so I treated the single supplied path (HTAV → architectural/spatial H&E features → AR-targeted therapy response) as the core mechanistic backbone and built the hypothesis from that path. 3) Tier 2 supporting evidence (used to justify mechanism but not included as anchors): literature linking morphologic heterogeneity to clonal diversity and adaptive resistance, and studies showing spatially structured microenvironments modulate drug selection pressures. 4) Endpoint choice: I selected therapy_response (AR-targeted therapy response) because the clinical decision problem is to predict sensitivity versus resistance to enzalutamide/abiraterone in the localized high-risk setting — a direct, actionable outcome for HTAV. 5) Deliberate exclusions: I excluded molecular anchor genes because none were provided by the ontologist/input and the task forbids inventing genes; I also excluded survival (OS/PFI) to keep the hypothesis focused on the immediate treatment-response mechanism and its clinical consequence?

Working Hypothesis

High HTAV indicates spatially structured tumor architecture that promotes AR-pathway–independent subclones, which are selected or enriched under AR blockade and drive adaptive resistance. Clinically, tumors with high HTAV will exhibit poorer response to AR-targeted therapy (e.g., lower biochemical/radiographic responses and shorter time-to-progression) compared with low-HTAV tumors, independent of standard clinical covariates.

Element	Value
Imaging feature	Histology-based Tumor Architectural Variability (HTAV) score
Primary endpoint	therapy_response

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: Retrospective/prospective cohorts with multi-region tumor sampling allow HTAV scoring from diagnostic/separator regions and linkage to AR-targeted therapy response metrics (PSA decline, radiographic response, time-to-progression).

Prediction 2: Statistical: multivariable logistic regression for response; Cox proportional hazards models for time-to-progression, adjusting for Gleason score, tumor stage, PSA, treatment modality, and other covariates.

Prediction 3: Mechanistic support: spatial transcriptomics across regions to derive AR signaling activity scores and assess regional heterogeneity in AR pathway activity; region-specific ex vivo or organoid assays under AR blockade to evaluate differential survival, without naming anchor genes.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions